

$$\Leftrightarrow P \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = \vec{0} \Leftrightarrow \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = P^{-1} \vec{0} = \vec{0}.$$

(Review this first).

Then by our result about matrix representation under change of

basis, we have: $[T]_B = P [T]_{B'} P^{-1} = U$

$\Rightarrow [T]_B$ is an upper triangular matrix.

Notation: Let $g(x) = a_0 + a_1x + \dots + a_mx^m \in IP(F)$. Then for any $\forall A \in F^{n \times n}$, then

$$g(A) = a_0I + a_1A + \dots + a_mA^m.$$

Remark: For any $g(x) = (x-a)(x-b)$ and any $A \in F^{n \times n}$

$$(A-aI)(A-bI) = (A-bI)(A-aI).$$

where $F = \mathbb{C}$ or $\mathbb{R}$.

Theorem (Cayley-Hamilton) Let $A \in F^{n \times n}$ and let $f(\lambda)$ be the characteristic polynomial of A . Then $f(A) = 0$.

Proof: For any $A \in F^{n \times n} \subseteq \mathbb{C}^{n \times n}$, there exists an upper triangular matrix

$$U = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} \in \mathbb{C}^{n \times n} \text{ and an invertible matrix } P \in \mathbb{C}^{n \times n} \text{ such}$$

that $A = PUP^{-1}$.

Thus 1). $f(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2) \dots (\lambda - \lambda_n)$

2). Denote $P = (\vec{p}_1 \vec{p}_2 \dots \vec{p}_n)$. Then $\{\vec{p}_1, \dots, \vec{p}_n\}$ forms a basis for $\mathbb{C}^n$.

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3). Denote $U = \begin{pmatrix} \lambda_1 & u_{12} & \dots & u_{1n} \\ & \lambda_2 & \dots & u_{2n} \\ & & \ddots & \vdots \\ & & & \lambda_n \end{pmatrix}$. Then

$$A = P U P^{-1} \Leftrightarrow AP = P U$$

$$\Leftrightarrow A(\vec{p}_1 \ \vec{p}_2 \ \dots \ \vec{p}_n) = (\vec{p}_1 \ \vec{p}_2 \ \dots \ \vec{p}_n) \begin{pmatrix} \lambda_1 & u_{12} & \dots & u_{1n} \\ & \ddots & & \vdots \\ & & \ddots & \\ & & & \lambda_n \end{pmatrix}$$

$$\Leftrightarrow (A\vec{p}_1 \ A\vec{p}_2 \ \dots \ A\vec{p}_n) = (\vec{p}_1 \ \vec{p}_2 \ \dots \ \vec{p}_n) \begin{pmatrix} \lambda_1 & * & \dots & * \\ & \ddots & & \\ & & \ddots & \\ & & & \lambda_n \end{pmatrix}$$

$$\Leftrightarrow \begin{cases} A\vec{p}_1 = \lambda_1 \vec{p}_1 & \Leftrightarrow (A - \lambda_1 I)\vec{p}_1 = \vec{0} & \textcircled{1} \\ A\vec{p}_2 = u_{12}\vec{p}_1 + \lambda_2 \vec{p}_2 & \Leftrightarrow (A - \lambda_2 I)\vec{p}_2 = u_{12}\vec{p}_1 & \textcircled{2} \\ \vdots & \vdots \\ A\vec{p}_k = u_{1k}\vec{p}_1 + u_{2k}\vec{p}_2 + \dots + u_{k-1,k}\vec{p}_{k-1} + \lambda_k \vec{p}_k & \Leftrightarrow (A - \lambda_k I)\vec{p}_k = \sum_{i=1}^{k-1} u_{ik}\vec{p}_i & \textcircled{k} \\ \vdots & \vdots \end{cases}$$

Claim: $\vec{p}_k \in \text{Null}[(A - \lambda_1 I) \dots (A - \lambda_k I)]$. for all $k=1, 2, \dots$

Remark: for any $m \leq k$: $\text{Null}[(A - \lambda_1 I) \dots (A - \lambda_m I)] \subseteq \text{Null}[(A - \lambda_1 I) \dots (A - \lambda_k I)]$

Proof of the claim by induction on k .

Base case: when $k=1$: $\vec{p}_1 \in \text{Null}(A - \lambda_1 I)$. by $\textcircled{1}$ $(A - \lambda_1 I)\vec{p}_1 = \vec{0}$

Inductive steps: Suppose $\vec{p}_{i-1} \in \text{Null}[(A - \lambda_1 I) \dots (A - \lambda_{i-1} I)]$. for all $i=1, \dots, k$

$$\begin{aligned} \text{Then as } (A - \lambda_k I)\vec{p}_k &= \sum_{i=1}^{k-1} u_{ik}\vec{p}_i && \text{Null}[(A - \lambda_1 I) \dots (A - \lambda_{k-1} I)] \\ &= u_{1k}\vec{p}_1 + u_{2k}\vec{p}_2 + \dots + \dots + u_{k-1,k}\vec{p}_{k-1} && \uparrow \\ &&& \text{Null}[(A - \lambda_1 I)(A - \lambda_2 I) \dots (A - \lambda_{k-1} I)] \\ &\in \text{Null}[(A - \lambda_1 I)(A - \lambda_2 I) \dots (A - \lambda_{k-1} I)] \end{aligned}$$

$$(A - \lambda_k I)\vec{p}_k \in \text{Null}[(A - \lambda_1 I)(A - \lambda_2 I) \dots (A - \lambda_k I)] \Rightarrow \vec{p}_k$$

$$\Rightarrow (A - \lambda_1 I)(A - \lambda_2 I) \cdots (A - \lambda_n I) \left[(A - \lambda_k I) \vec{p}_k \right] = \vec{0}$$

$$\Rightarrow \vec{p}_k \in \text{Nul} \left[(A - \lambda_1 I)(A - \lambda_2 I) \cdots (A - \lambda_k I) \right]$$

$$\Rightarrow \text{By the remark 1: } \vec{p}_i \in \text{Nul} \left[(A - \lambda_1 I) \cdots (A - \lambda_i I) \right] \text{ for all } i=1, \dots, n$$

$$\Rightarrow f(A)(\vec{p}_i) = \vec{0} \text{ for all } i=1, \dots, n.$$

$$\text{As } \{\vec{p}_1, \dots, \vec{p}_n\} \text{ is a basis of } \mathbb{C}^n, f(A) = 0.$$

Corollary. Let T be a linear transformation over finite dimensional vector space V . Then $f(\lambda) = \det(T - \lambda I)$ be the characteristic polynomial of T . Then $f(T) = 0$.

Proof: Let $\mathcal{B}$ be any ordered basis for V .

$$\text{Then } f(\lambda) = \det([T]_{\mathcal{B}} - \lambda I)$$

$$\text{Then } f([T]_{\mathcal{B}}) = 0$$

$$\Rightarrow [f(T)]_{\mathcal{B}} = 0 \quad (\text{by linearity of } T \text{ and } [T^k]_{\mathcal{B}} = ([T]_{\mathcal{B}})^k)$$

$$\Rightarrow f(T) = 0.$$

Inner Products.

Definition: Let V be a vector space over $\mathbb{C}$. An inner product on V is a complex-valued function of two ordered vectors in V , denoted by $(\cdot, \cdot)$, which satisfies the following conditions.

- (I) $(\vec{x}, \vec{x}) \geq 0$ for all $\vec{x} \in V$ and $(\vec{x}, \vec{x}) = 0$ if and only if $\vec{x} = \vec{0}$.
- (II) $(\vec{x}, \vec{y}) = \overline{(\vec{y}, \vec{x})}$ for all $\vec{x}, \vec{y} \in V$.
- (III) $(c\vec{x}, \vec{y}) = c(\vec{x}, \vec{y})$ for all $\vec{x}, \vec{y} \in V$ and $c \in \mathbb{C}$.
- (IV) $(\vec{x} + \vec{y}, \vec{z}) = (\vec{x}, \vec{z}) + (\vec{y}, \vec{z})$ for all $\vec{x}, \vec{y}$, and $\vec{z} \in V$.

Remarks: $(\vec{x}, c\vec{y}) = \overline{(c\vec{y}, \vec{x})} = \overline{c(\vec{y}, \vec{x})} = \bar{c} \overline{(\vec{y}, \vec{x})} = \bar{c}(\vec{x}, \vec{y})$

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Definition: Let V be a vector space over $\mathbb{R}$. An inner product on V is a **real**-valued function of two ordered vectors in V , denoted by $(\cdot, \cdot)$, which satisfies the following conditions.

- (I) $(\vec{x}, \vec{x}) \geq 0$ for all $\vec{x} \in V$ and $(\vec{x}, \vec{x}) = 0$ if and only if $\vec{x} = \vec{0}$.
- (II) $(\vec{x}, \vec{y}) = (\vec{y}, \vec{x})$ for all $\vec{x}, \vec{y} \in V$.
- (III) $(c\vec{x}, \vec{y}) = c(\vec{x}, \vec{y})$ for all $\vec{x}, \vec{y} \in V$ and $c \in \mathbb{R}$.
- (IV) $(\vec{x} + \vec{y}, \vec{z}) = (\vec{x}, \vec{z}) + (\vec{y}, \vec{z})$ for all $\vec{x}, \vec{y}$, and $\vec{z} \in V$.

Definition. A vector space V over F endowed with a specific inner product is called an inner product space. If $F = \mathbb{C}$, we call V a complex inner product space. If $F = \mathbb{R}$, we call V a real inner product space.

Examples: 1) Let $V = \mathbb{R}^n$. $\vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$, $\vec{y} \in \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$ define $(\vec{x}, \vec{y}) = \vec{x}^T \vec{y} = (x_1 \ x_2 \ \dots \ x_n) \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$
 $= x_1 y_1 + x_2 y_2 + \dots + x_n y_n.$

2) Let $V = \mathbb{C}^n$. $\vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$, $\vec{y} \in \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$ define: $(\vec{x}, \vec{y}) = \vec{y}^* \vec{x} = (x_1 \ x_2 \ \dots \ x_n) \begin{pmatrix} \bar{y}_1 \\ \bar{y}_2 \\ \vdots \\ \bar{y}_n \end{pmatrix}$
 $= x_1 \bar{y}_1 + x_2 \bar{y}_2 + \dots + x_n \bar{y}_n$
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The above inner products on $\mathbb{R}^n$ or $\mathbb{C}^n$ are referred as standard inner product

3) Let $V = \mathbb{R}^{n \times n}$. define $(A, B) = \text{Tr}(A^T B)$.

Check: i). $(A, A) = \text{Tr}(A^T A) \geq 0$ b/c $\text{Tr}(A^T A) = \sum_{j=1}^n \sum_{i=1}^n a_{ij}^2 \geq 0$.

and $(A, A) = 0$ if and only if $A = 0$. (Homework problem result)

II). $(A, B) = \text{Tr}(A^T B) = \text{Tr}(\underbrace{(A^T B)^T}) = \text{Tr}(B^T A) = (B, A)$

b/c Trace of a matrix equals to the trace of its transpose.

III). $(cA, B) = \text{Tr}((cA)^T B) = \text{Tr}(c A^T B) = c \text{Tr}(A^T B) = c(A, B)$

IV) $(A+B, C) = \text{Tr}((A+B)^T C) = \text{Tr}(A^T + B^T)C = \text{Tr}(A^T C + B^T C)$
 $= \text{Tr}(A^T C) + \text{Tr}(B^T C) = (A, C) + (B, C).$

4). Let $V = \mathbb{C}^{n \times n}$. Check that $(A, B) = \text{Tr}(B^* A)$ defines an inner product on $\mathbb{C}^{n \times n}$.

5). Let $V = C^0([-1, 1])$ be the vector space of the set of continuous real-valued functions on $[-1, 1]$. Define

$$(f, g) = \int_{-1}^1 f(x) g(x) dx.$$

Check: i). $(f, f) = \int_{-1}^1 f(x)^2 dx \geq 0$. and $(f, f) = 0$ if and only if $f = 0$.

II) $(f, g) = \int_{-1}^1 f(x) g(x) dx = \int_{-1}^1 g(x) f(x) dx = (g, f)$

III) $(cf, g) = \int_{-1}^1 cf(x) g(x) dx = c \int_{-1}^1 f(x) g(x) dx = c(f, g)$